FPT UNIVERSITY — HO CHI MINH CITY CAMPUS
DEPARTMENT OF SEMICONDUCTOR AND AUTOMOTIVE ENGINEERING

FPT UNIVERSITY

LAB REPORT

Lab 4 — Sequential MOS Logic Circuit

CMOS D Flip-Flop Design

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Class	IC2007
Subject	ICI211

1 BACKGROUND THEORY

A D flip-flop is a sequential circuit that captures the input data D at a specific clock edge and holds it at the output Q. The output changes only on the triggering edge: on a rising edge Q takes the value of D, and between edges the stored value is held.

It is built from two latches: a master latch driven by CLK (active while CLK = 1) and a slave latch driven by the inverted clock (active while CLK = 0). Only one latch is transparent at a time, which is what makes the overall cell edge-triggered rather than level-sensitive.

CLK edge	D	Q_{next}
Rising	0	0
Rising	1	1
Falling	0	Hold
Falling	1	Hold

1.1 Timing constraints

Three parameters characterise the temporal behaviour of the cell. Setup time is the minimum interval the data must stay stable before the clock edge; hold time is the minimum interval it must stay stable after the edge; clock-to-output delay is the time the output needs to respond once the edge has occurred.

Symbol	Definition
t_{setup}	Data stable before the clock edge
t_{hold}	Data stable after the clock edge
$t_{clk→Q}$	Clock-to-output delay

$$t_{setup} + t_{hold} \leq T_{clk} \qquad t_{clk \to Q} = t_{pd}$$

2 PART A — SCHEMATIC DESIGN

The flip-flop is assembled from CMOS transmission gates (pass gates) and inverters. Each cell was drawn at transistor level and wrapped in a symbol for instantiation at the top level.

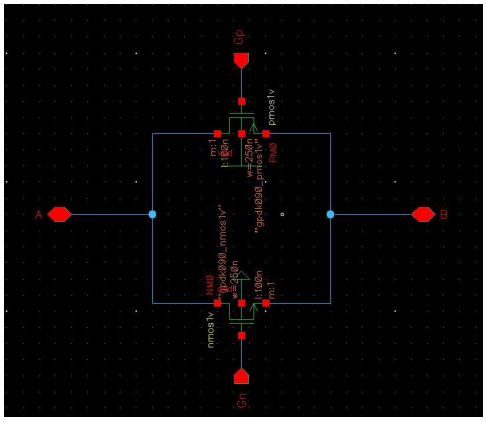


Figure 1. *Schematic of the pass gate.*

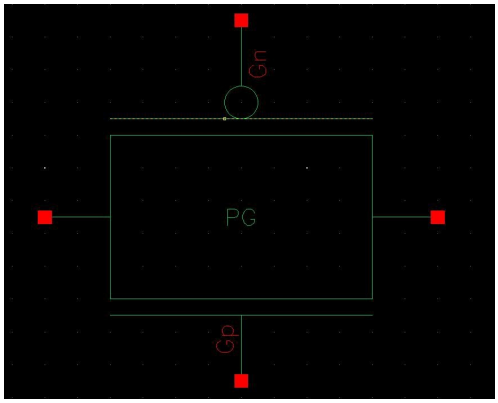


Figure 2. *Symbol of the pass gate.*

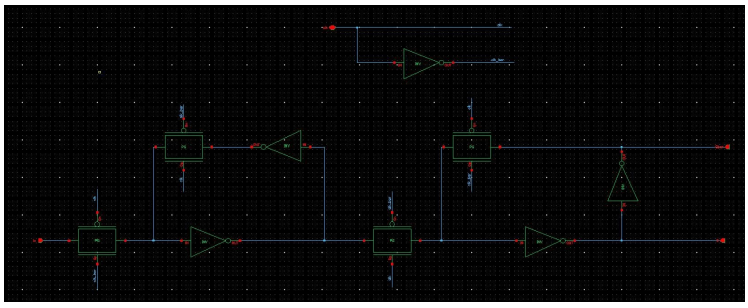


Figure 3. *Schematic of the CMOS D flip-flop.*

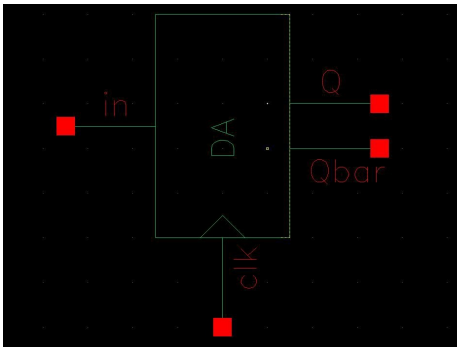


Figure 4. *Symbol of the D flip-flop.*

3 PRE-LAYOUT SIMULATION

3.1 Functional verification

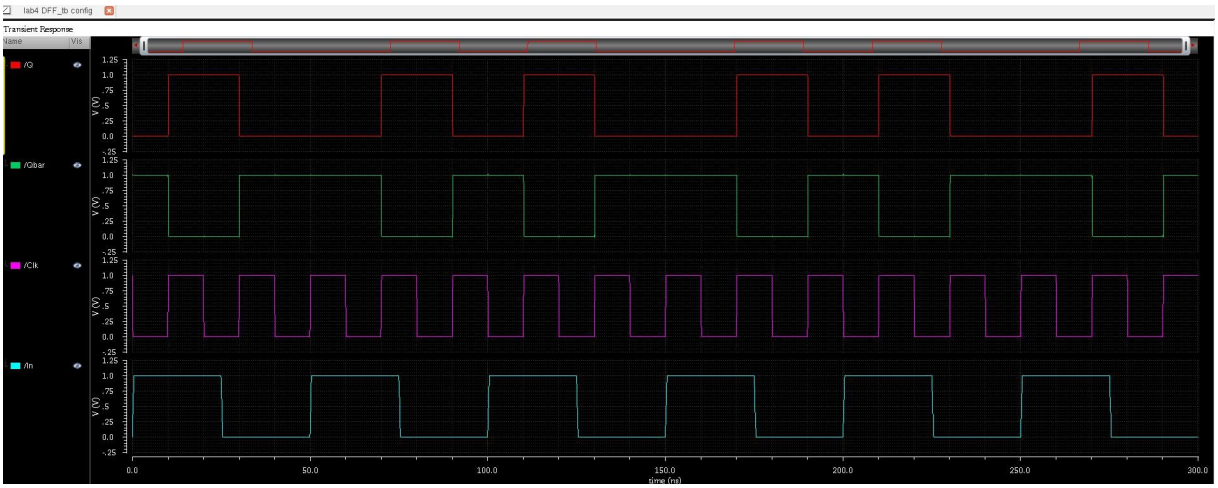


Figure 5. *Pre-layout transient simulation of the CMOS D flip-flop.*

> **Conclusion:** Q follows D only on rising clock edges and holds its value on falling edges — the pre-layout schematic behaves as an edge-triggered flip-flop.

3.2 Output Q — transition and propagation delay

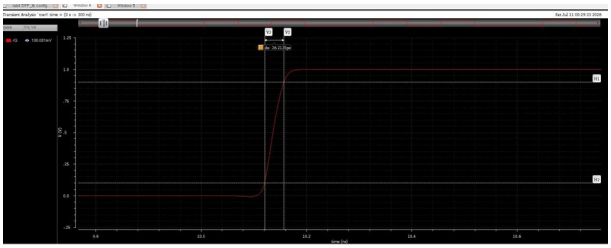


Figure 6. *Q rise time t_r = 36.2135 ps.*

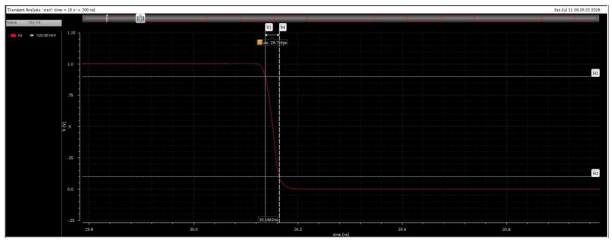


Figure 7. *Q fall time t_f = 26.724 ps.*

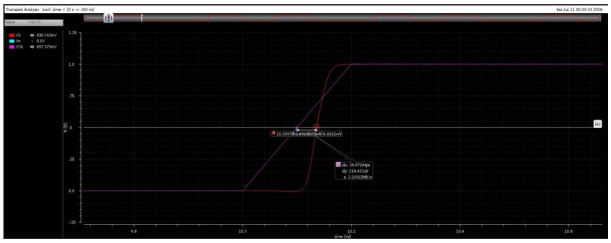


Figure 8. *Q propagation delay t_{pLH} = 36.67 ps.*

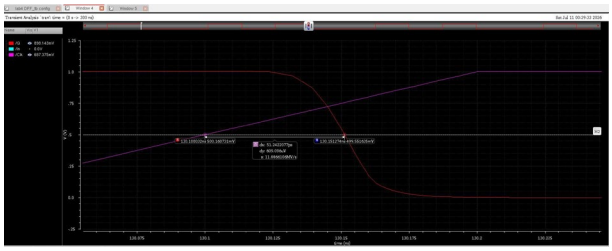


Figure 9. *Q propagation delay t_{pHL} = 51.24 ps.*

$$T_p = (t_{pHL} + t_{pLH}) / 2 = (51.24 + 36.67) / 2 = 43.955\ ps$$

3.3 Output Qbar — transition and propagation delay

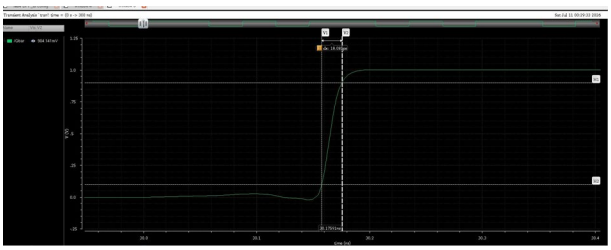


Figure 10. *Qbar rise time t_r = 18.81 ps.*

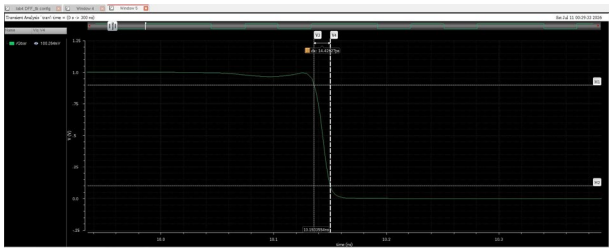
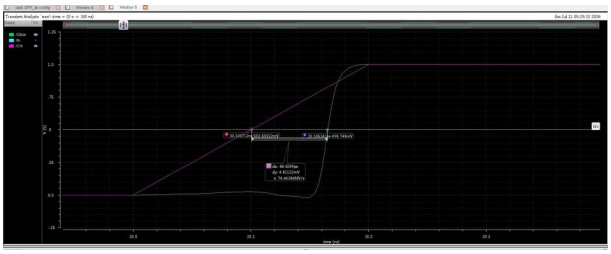


Figure 11. *Qbar fall time t_f = 14.42 ps.*



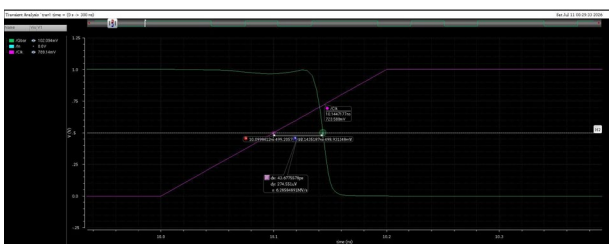


Figure 12. *Qbar propagation delay t_{pLH} = 64.63 ps.*

Figure 13. *Qbar propagation delay t_{pHL} = 43.67 ps.*

$$T_p = (t_{pHL} + t_{pLH}) / 2 = (64.63 + 43.67) / 2 = 54.15\ ps$$

3.4 Clock-to-output delay summary

Table 1 — Extracted timing of the CMOS D flip-flop (pre-layout)

Output	t_r (ps)	t_f (ps)	t_{pLH} (ps)	t_{pHL} (ps)	T_p (ps)
Q	36.2135	26.724	36.67	51.24	**43.955**
Qbar	18.81	14.42	64.63	43.67	**54.15**

Conclusion: All delays are measured at the 50 % threshold of the supply voltage (V_{DD}/2), from the CLK edge to the corresponding output.

3.5 Setup time characterization

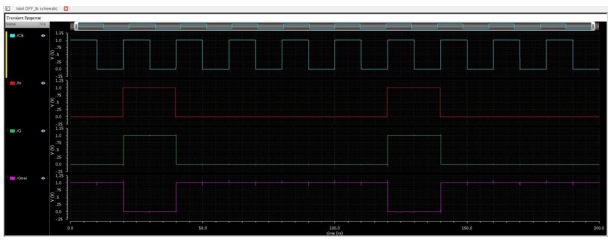


Figure 14. *Setup time non-violation.*

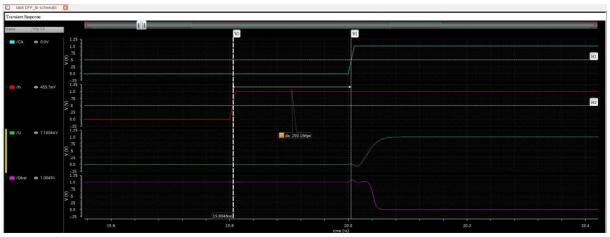


Figure 15. *Setup time non-violation (closer look).*

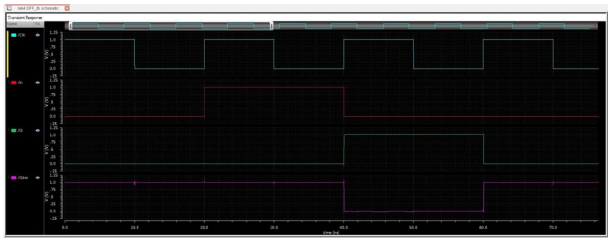


Figure 16. *Setup time violation.*

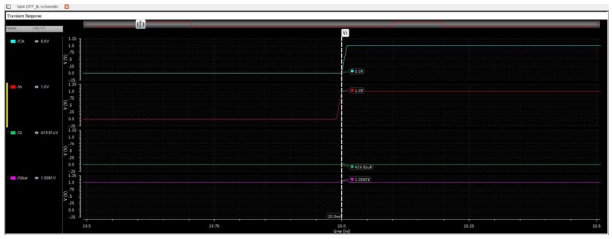


Figure 17. *Setup time violation (closer look).*

Table 2 — Setup time measurement

Parameter	Measured value	Condition / threshold
Setup time (t_{setup})	$t_{setup} = T_{CLK} - T_D$ = 20 ns − 19.95 ns = **0.05 ns**	The point where the nominal T_p degrades (increases) by 10 % due to metastability.

Conclusion: With the triggering clock edge fixed at 20.00 ns and the data arrival progressively delayed, metastability appears once D transitions at 19.95 ns: the master latch can no longer capture the logic state before the transmission gate turns off, because its internal parasitic capacitances have not charged to the correct level. The minimum setup time is therefore t_{setup} = 20.00 ns − 19.95 ns = 0.05 ns (50 ps).

3.6 Hold time characterization

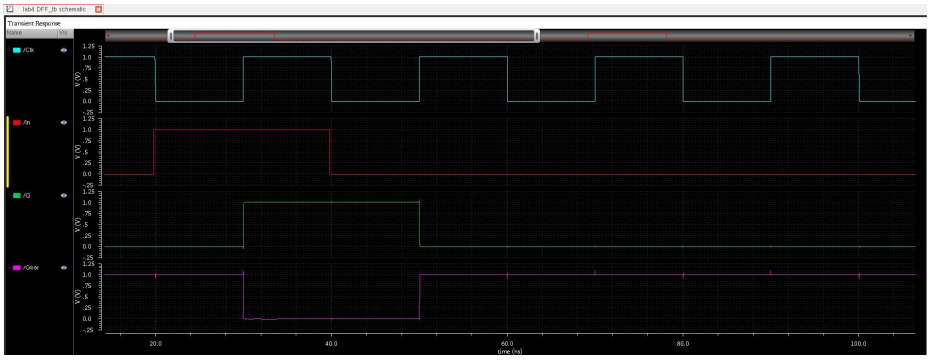


Figure 18. *Hold time non-violation.*

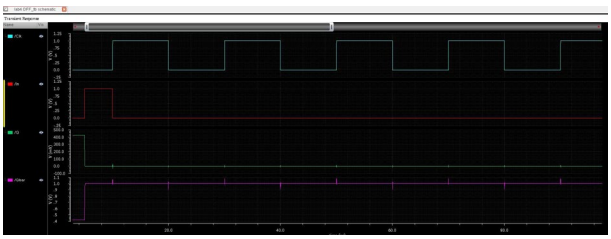


Figure 19. *Hold time violation.*

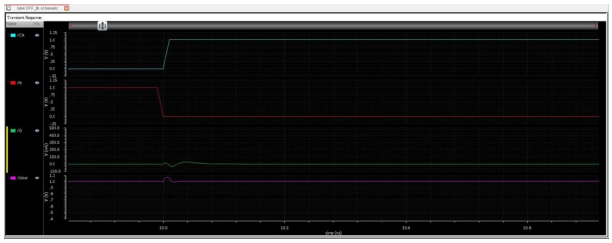


Figure 20. *Hold time violation (closer look).*

Table 3 — Hold time measurement

Parameter	Measured value	Condition / threshold
Hold time (t_{hold})	$t_{hold} = T_D - T_{CLK}$ = 9.99 ns − 10 ns = **−0.01 ns**	The point where the nominal T_p degrades by 10 %, or the flip-flop fails to capture data.

Conclusion: With D rising at 5.00 ns and the clock edge at 10.00 ns, a data pulse of 4.98 ns (D falling at 9.98 ns) caused a complete latch failure, while 4.99 ns (D falling at 9.99 ns) was marginally stable — giving t_{hold} = 9.99 ns − 10.00 ns = −0.01 ns (−10 ps). The negative hold time comes from the internal clock buffering (the inverters generating clk and clk_bar), whose propagation delay lets the data be removed 10 ps before the external clock edge arrives — a robust characteristic of this pass-gate architecture.

4 PART B — LAYOUT DESIGN

The pass gate and the complete flip-flop were laid out with compact routing and shared diffusion, then signed off with DRC and LVS.

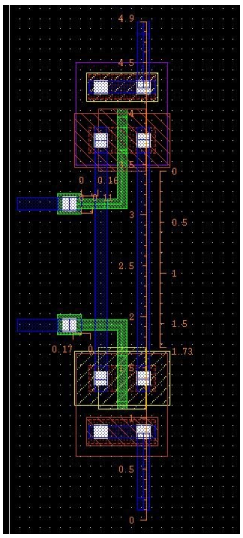


Figure 21. *Pass-gate layout.*

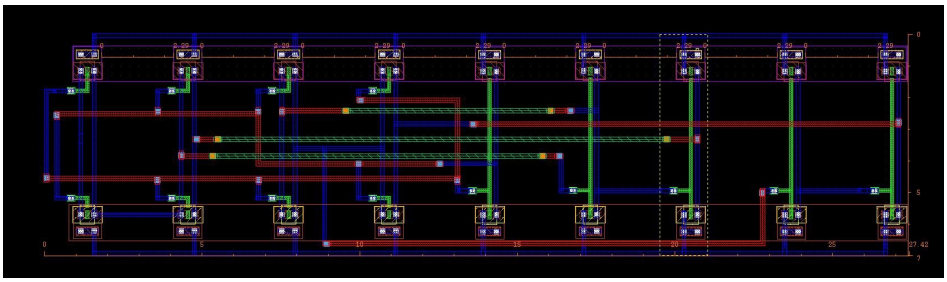


Figure 22. *D flip-flop layout.*

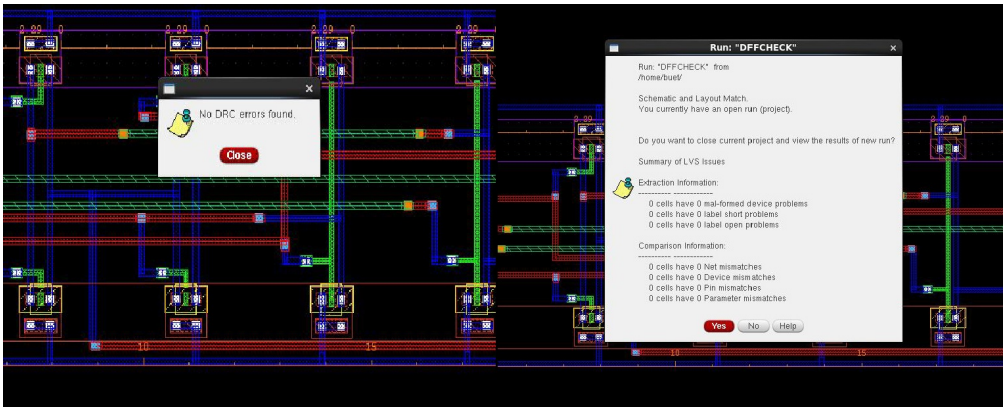


Figure 23. *Design Rule Check (DRC) and Layout Versus Schematic (LVS) results.*

Conclusion: The layout is DRC clean (0 errors) and LVS matched — connectivity and device sizing are identical to the schematic. Compact routing and diffusion sharing keep the cell area and parasitic footprint to a minimum.

5 PART C — POST-LAYOUT SIMULATION

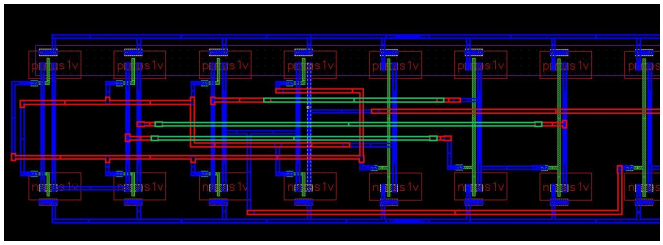


Figure 24. *Post-layout extracted view of the DFF.*

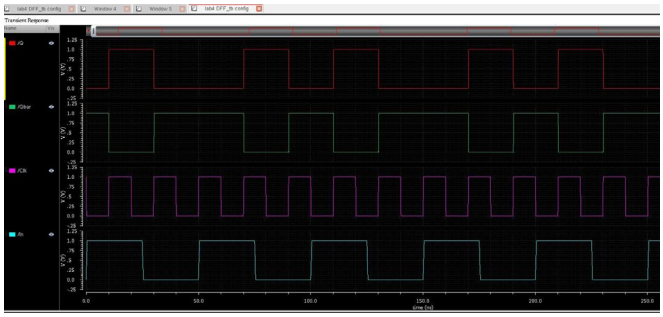


Figure 25. *Post-layout simulation of the DFF.*

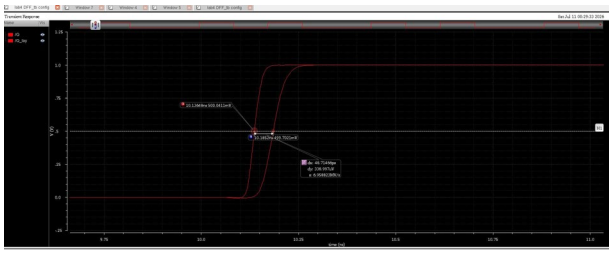


Figure 26. *Pre- vs. post-layout delay of Q.*

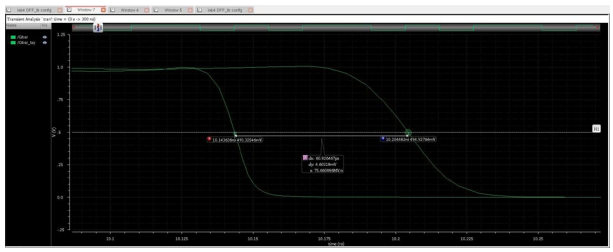


Figure 27. *Pre- vs. post-layout delay of Qbar.*

Conclusion: Figures 26 & 27 show a clear divergence between the pre- and post-layout waveforms: both Q and Qbar are noticeably slower after extraction. The added delay is driven by the parasitic RC of the physical interconnect, routing layers and shared diffusion regions, confirming that careful layout optimization is indispensable in high-frequency CMOS sequential design.

6 EXERCISE 1 — BASIC LATCH VERIFICATION

Level-sensitive latches and their edge-triggered flip-flop counterparts were designed and simulated to contrast the two triggering mechanisms.

6.1 SR latch and SR flip-flop

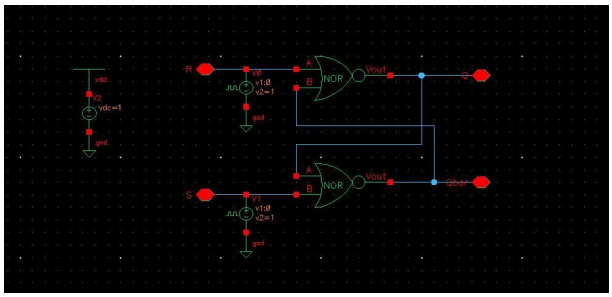


Figure 28. *SR latch schematic.*

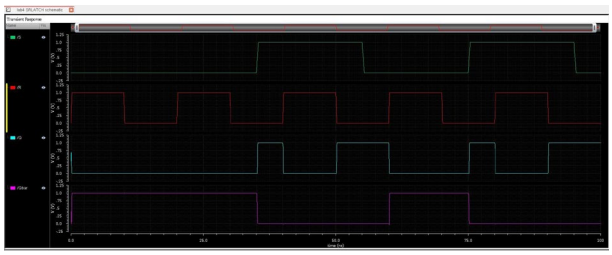


Figure 29. *SR latch simulation.*

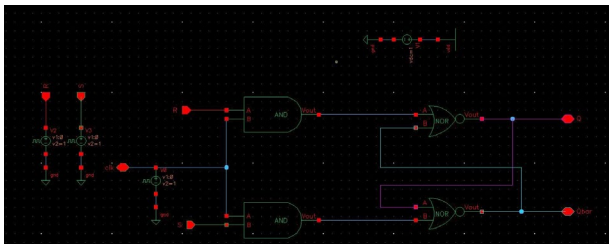


Figure 30. *SR flip-flop schematic.*

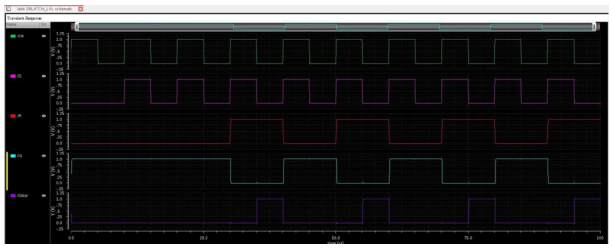


Figure 31. *SR flip-flop simulation.*

Conclusion: The simulations confirm the level-sensitive nature of the SR latch: while CK is low the latch holds its previous state and the inputs have no influence; while CK is high the inputs propagate through the NOR gates and the outputs follow the Set/Reset conditions. When the clock returns to zero the current state is locked and maintained.

6.2 T latch and T flip-flop

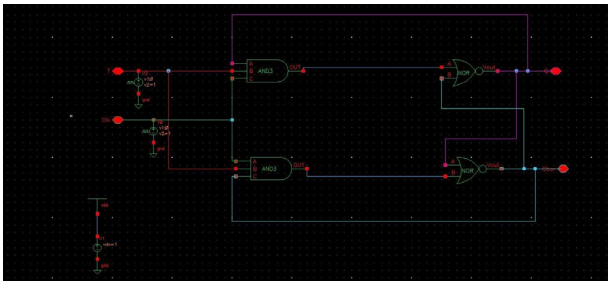


Figure 32. *T latch schematic.*

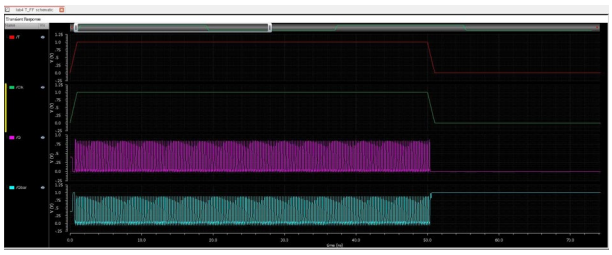


Figure 33. *T latch simulation.*

Table 4 — T latch verification (level-sensitive)

Enable (E)	T	Q_{next}	Description
0	X	Q	Holds previous state
1	0	Q	Holds current state
1	1	¬Q	Race-around condition (continuous toggling)

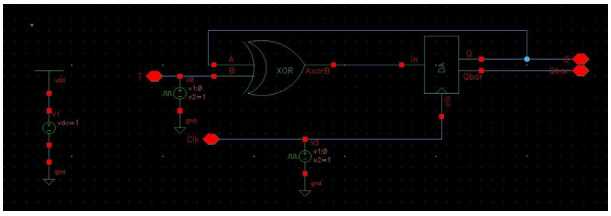


Figure 34. *T flip-flop schematic.*

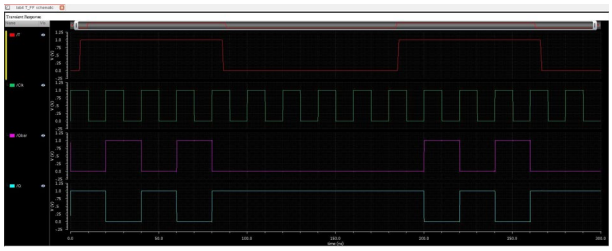


Figure 35. *T flip-flop simulation.*

Table 5 — T flip-flop verification (edge-triggered)

T (toggle)	Q (current)	D input (T ⊕ Q)	Q_{next} (after edge)	Circuit behaviour
0	0	0 ⊕ 0 = 0	0	Hold state
0	1	0 ⊕ 1 = 1	1	Hold state
1	0	1 ⊕ 0 = 1	1	Toggle state
1	1	1 ⊕ 1 = 0	0	Toggle state

Conclusion: The two circuits illustrate the difference between level-sensitive and edge-triggered operation. In the T latch, holding Enable high while T = 1 makes the output toggle continuously — an uncontrolled race-around condition. The edge-triggered T flip-flop removes this instability: it toggles exactly once per clock edge when T = 1 and holds its state otherwise, giving predictable synchronous behaviour.

7 EXERCISE 2 — REGISTER DESIGN EXTENSION

7.1 D flip-flop with asynchronous reset

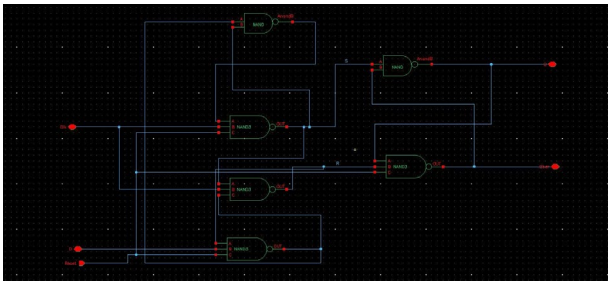


Figure 36. *D flip-flop with asynchronous reset — (a) schematic, (b) symbol.*

The reset input clears the stored value independently of the clock, and is the building block reused in both registers below.

7.2 4-bit parallel register

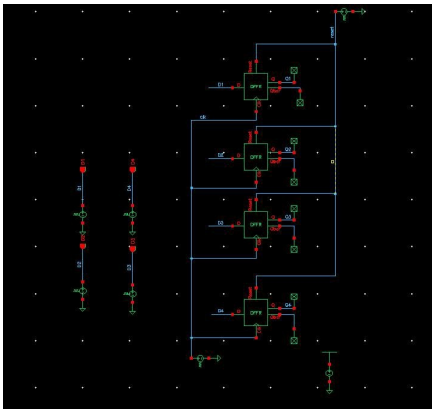


Figure 37. *4-bit register schematic.*



Figure 38. *Simulation while waiting for the rising clock edge.*



Figure 39. *At the rising clock edge — all Qs rose.*

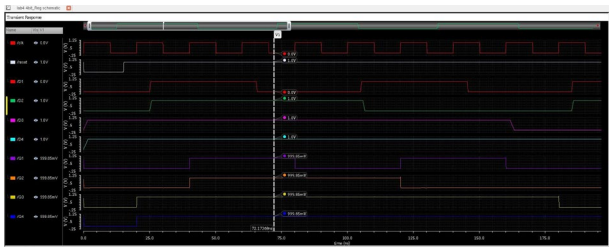


Figure 40. *Holding when CLK falls (positive-edge operation).*



Figure 41. *At the rising clock edge — all Qs fell.*

Conclusion: The 4-bit register performs parallel data storage together with asynchronous reset. Asserting the active-low reset immediately clears all output bits (Q1–Q4) regardless of the clock state; during normal synchronous operation the register captures and holds the parallel inputs D1–D4 on every triggering clock edge, acting as a stable multi-bit storage unit.

7.3 Shift register (SIPO)

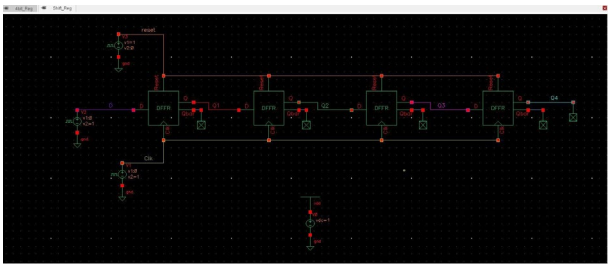


Figure 42. *Shift register schematic.*

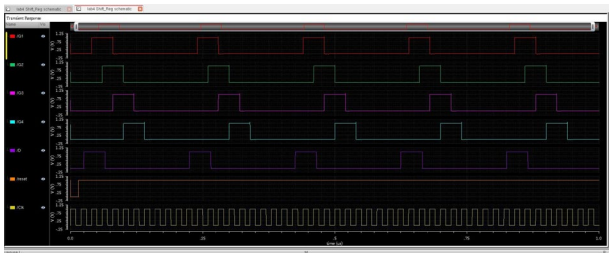


Figure 43. *Shift register simulation.*

Conclusion: Cascading the D flip-flops produces the expected shifting behavior: data applied at the serial input D propagates to the first stage Q1 on the initial clock edge, and on each subsequent cycle the stored bit moves to the next stage (Q1 → Q2 → Q3 → Q4) — a correct Serial-In Parallel-Out transfer.